

Maruf Adesina Lawal

Curriculum Vitæ

Department of Mathematics
University of Tennessee
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Research Interests

- Mathematical Biology and Ecology, Infectious Diseases Modeling, Natural Resources Modeling, Optimal Control Theory, Machine Learning/AI for Health, Data-Driven Modeling of Biological Systems, Data Analysis, Computational Mathematics.

Education

- May. 2026 (Projected) **PhD, Mathematics, Department of Mathematics, University of Tennessee (UTK)**, Knoxville, USA
Dissertation (in progress): *Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy and Optimal Sustainable Management of Multi-Species Harvesting in the Southern Black Sea Ecosystem*
Advisor: [Prof. Suzanne Lenhart](#)
- May. 2023 **Master of Science, Mathematics, Department of Mathematics, University of Tennessee (UTK)**, Knoxville, USA
- Feb. 2015 **B.Tech, Mathematics, Department of Pure and Applied Mathematics, Ladoke Akintola University of Technology (LAUTECH)**, Ogbomoso, Nigeria
Distinction : *first class honours*
Thesis : *Mathematical modeling of a standard incidence epidemiological model with relapse*
Advisor: Dr. Samson Olaniyi

Professional Experience

- 2022 - Current **Graduate Teaching Associate, University of Tennessee (UTK)**, Knoxville, Tennessee, USA
- 2021 - 2022 **Graduate Teaching Assistant, University of Tennessee (UTK)**, Knoxville, Tennessee, USA
- 2019 - 2021 **High School Mathematics Teacher, Learn-With-Ease (LWE) Schools**, Lagos, Nigeria
- 2018 - 2021 **Tutor, Tractrain Consulting**, Lagos, Nigeria
- 2015 - 2017 **Teaching Assistant, Federal Polytechnic Kaura Namoda (FPKN)**, Zamfara, Nigeria

Publications

C. W. Chukwu, [Maruf A. Lawal](#), D. Aldila, and G. Obaido, Evaluating Syphilis Transmission Among MSM Population: A Mathematical Model for Reinfection, Two-Stage Treatment and Treatment Failure. *Front. Public Health* **14**:1841745, 2026. [doi:10.3389/fpubh.2026.1841745](https://doi.org/10.3389/fpubh.2026.1841745).

[M. A. Lawal.](#), C. W. Chukwu, K. R. Cheneke, and D. Wanduku, Simulation-Based Predictive Modeling of Behavioral Heterogeneity in Chlamydia Transmission: Insights from U.S. Surveillance Data (*Under Review*).

[M. A. Lawal](#), M. Demir, M. J. Kelly, and S. Lenhart, Optimal Sustainable Management of Multi-Species Harvesting in the Southern Black Sea Ecosystem (*Under Review*).

Samson Olaniyi, [Maruf A. Lawal](#) and Olawale S. Obabiyi, Stability and Sensitivity Analysis of a Deterministic Epidemiological Model with Pseudo-recovery, *IAENG International Journal of Applied Mathematics*, vol. 46, No 2, Pages 160-167, 2016.

[M. A. Lawal.](#), C. Edholm, H. Correia, F. Augusto, S. Lenhart, W. Chukwu, D. Goldsman, F. Chirove, J. White, and S. Dobson, Modeling the Interplay of Vaccine Hesitancy and Disease Dynamics (Manuscript in preparation).

[M. A. Lawal.](#), Mufutau R. A., Gbadamosi A. A., Adebiyi A. A., and Olayinka H. A, Mathematical Modeling of Tuberculosis: A Comprehensive Review (Manuscript in preparation).

Farai Chirove, [M. Lawal.](#), Suzanne Lenhart, Zhisheng Shuai, Munkaila Dasumani, and Edward Lungu, Within-Host MPXV Model with Immune Response, Antiviral Therapy, and Drug Toxicity (Manuscript in preparation).

Invited Talks

- Mar. 2026 **Modeling the Interplay of Vaccine Hesitancy and Disease Dynamics**, *Spring 2026 AMS Southeastern Sectional Meeting*, Savannah, GA, USA
- Jan. 2026 **Optimal Sustainable Management of Multi-Species Harvesting in the Southern Black Sea Ecosystem**, *2026 Joint Mathematics Meetings (JMM)*, Washington, DC, USA
- March. 2025 **Optimal Sustainable Management of the Horse Mackerel Harvesting in the Southern Black Sea Coast**, *2025 SIAM Southeastern Atlantic Section Meeting*, Knoxville, TN, USA
- Jan. 2025 **Optimal Sustainable Management of the Horse Mackerel Harvesting in the Southern Black Sea Coast**, *2025 Joint Mathematics Meetings (JMM)*, Seattle, WA, USA
- March 2024 **Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy**, *103rd Annual MAA Southeastern Sectional Meeting*, Knoxville, TN, USA
- Jan. 2024 **Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy**, *2024 Joint Mathematics Meetings (JMM)*, San Francisco, CA, USA

Contributed Talks

- June 2026 **Modeling the Interplay of Vaccine Hesitancy and Disease Dynamics**, *Modeling the Invisible Workshop*, Georgia State University, Atlanta, GA, USA
- Nov. 2024 **Optimal Sustainable Management of the Horse Mackerel Harvesting in the Southern Black Sea Coast**, *Seminar: Mathematics Presentations*, University of Tennessee, Knoxville, TN, USA
- Sept. 2024 **Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy**, *Seminar: Mathematics Presentations*, University of Tennessee, Knoxville, TN, USA

Conferences and Workshops Attended

- June 2026 **Modeling the Invisible Workshop**, Atlanta, GA, USA
- Mar. 2026 **2026 Spring AMS Southeastern Sectional Meeting**, Savannah, GA, USA
- Jan. 2026 **2026 Joint Mathematics Meetings (JMM)**, Washington, DC, USA
- Nov. 2025 **Masamu Advanced Study Institute (MASI) Convening Research Workshop (Virtual)**
- Nov. 2025 **2025 Virtual SAMSA Masamu Advanced Study Institute (MASI) Convening**
- June/July 2025 **2025 Institute for Mathematics and its Applications (IMA) Math-to-Industry Boot Camp**, virtual
Project: Proactive Housing Support: using public data to predict and prevent evictions
- June. 2025 **2025 Institute for Mathematical and Statistical Innovation (IMSI) Data Science Boot Camp**, virtual
- April. 2025 **2025 One Health Student Leadership Summit**, University of Tennessee, Knoxville, TN, USA
- March. 2025 **2025 SIAM Southeastern Atlantic Section Meeting**, University of Tennessee, Knoxville, TN, USA
- Jan. 2025 **2025 Joint Mathematics Meetings (JMM)**, Seattle, WA, USA
- March 2024 **103rd Annual Mathematical Association of America (MAA) Southeastern Sectional Meeting**, University of Tennessee, Knoxville, TN, USA
- Jan. 2024 **2024 Joint Mathematics Meetings (JMM)**, San Francisco, CA, USA
- July 2023 **Society for Mathematical Biology Annual Meeting**, Ohio State University, Ohio, OH, USA

- June 2022 **Mathematical Problems in Industry (MPI) Workshop**, *Worcester Polytechnic Institute, MA, USA*
Project: Patient data generation and machine learning utilizing clinical data to predict the severity of heart failure
- June 2022 **Graduate Student Mathematical Modeling Camp (GSMMC)**, *University of Delaware, DE, USA*
Project: Asymmetric division and modeling of interacting cell populations in the colonic crypt
- Nov. 2022 **Twelfth Annual Masamu Advanced Study Institute (MASI) and Research Workshop**, *as part of the 2021 Southern African Mathematical Sciences Association (SAMSA) Conference held in Maputo, Mozambique (Virtual), Online*
- Nov. 2021 **Eleventh Annual Masamu Advanced Study Institute (MASI) and Research Workshop**, *as part of the 2021 Southern African Mathematical Sciences Association (SAMSA) Conference, Online*

Journal Review Work

- Scientific Reports (SR) Springer Nature
- Mathematical Modelling and Applications

Awards and Fellowships

- Winner, “Modeling the Invisible: Competition on Forecasting Viral Spread with Limited Data,” Modeling the Invisible Workshop June 2026
- FY23 Graduate School Fellowship (\$1018) Jan 2026
- University of Tennessee Graduate Student Senate (GSS) Travel Award (\$800) Jan. 2026
- FY23 Graduate School Fellowship (\$1018) Aug 2025
- FY22 University of Tennessee Graduate School Fellowship (\$2000) 2025
- AMS BEGIN Bootcamp Travel Grant (\$300) Jan. 2025
- 2025 AMS Joint Mathematics Meetings (JMM) Graduate Student Travel Grant (\$1430) Jan. 2025
- University of Tennessee Graduate Student Senate (GSS) Travel Award (\$500) Jan. 2025
- University of Tennessee Math Department Summer Dissertation Fellowship (\$4000) June 2024
- FY21 University of Tennessee Graduate School Fellowship (\$1249) 2024
- FY22 University of Tennessee Graduate School Fellowship (\$2000) 2024
- University of Tennessee Math Department Award (\$2300) 2024
- University of Tennessee Graduate Student Senate (GSS) Travel Award (\$650) Jan. 2024
- University of Tennessee Math Department Award (\$2000) 2023
- FY22 University of Tennessee Graduate School Fellowship (\$2000) 2023
- Yueh-er, Hong-hsu and Clarence Cheng Kuo Fellowship, at the 2023 UTK Mathematics Honors Day (\$1000) April 2023
- University of Tennessee Math Department Academic Excellence Fellowship (\$1660) 2022
- University of Tennessee Graduate School Award (\$400) 2022
- FY22 University of Tennessee Graduate School Fellowship (\$2000) 2022
- 2015 National Mathematics Competition for University Students (NAMCUS) Gold Medal Winner held at the National Mathematical Center, Abuja, Nigeria April 2015

Service

- Jan. 2025 **Reviewer** , *Undergraduate Poster Session, Joint Mathematics Meetings (JMM)*, Seattle, WA, USA
- 2024 - 2025 **Graduate Student Representative**, *University of Tennessee Technology Fee Advisory Board*
- 2024 - 2025 **Elected Senator**, *University of Tennessee Graduate Student Senate (GSS)*
- Jan. 2024 **Reviewer** , *Undergraduate Poster Session, Joint Mathematics Meetings (JMM)*, San Francisco, CA, USA
- Oct. 2023 **Panelist** , *University of Tennessee SIAM Graduate Student Internship/Workshop Experiences Panel*

Computing Proficiency

Programming **Python, MATLAB, R, Maple, $\LaTeX$**

Applications **Microsoft office Package (Word, Excel, Powerpoint)**

Memberships

- American Mathematical Society (AMS)
- Society for Mathematical Biology (SMB)
- Mathematical Association of America (MAA)
- Society for Industrial and Applied Mathematics (SIAM)
- Nigerian Mathematical Society (NMS)

Courses Taught at University Level

Instructor of Record

- Statistical Reasoning (MATH 115)–UTK Spring 2026
- Basic Calculus (MATH 125)–UTK Spring 2025
- Basic Calculus (MATH 125)–UTK Spring, Fall 2024
- Basic Calculus (MATH 125)–UTK Spring, Fall 2023
- Basic Calculus (MATH 125)–UTK Fall 2022
- Trigonometry and Analytic Geometry (MATH 122) – FPKN Semester 2, 2016

Co-Instructor

- Calculus I (MATH 141) – UTK Fall 2025
- Calculus I-B Infused with Precalculus (MATH 132) – UTK Fall 2025

Teaching Assistant

- Basic Calculus (MATH 125)–UTK Spring 2022
- College Algebra (MATH 119) – UTK Fall 2021

Online Courses/Certifications

- Complete Guide to R: Wrangling, Visualizing, and Modeling Data - *LinkedIn Learning* Oct, 2024
- Machine Learning with MATLAB - *MathWorks Training Services* May 2024
- Hands-On Advanced Python: Data Exploration and Manipulation - *LinkedIn Learning* May 2024
- Python for Data Science and Machine Learning Essential Training - *LinkedIn Learning* May 2024
- UTK Department of Mathematics GTA Teaching Advising Program May 2023